

IT³ Paradigm + MAGPIE v4.0.0 Polished — Unified Zero-Parameter Framework

> ****Zero free parameters. 12/12 modules verified. From CMB to Self-Sovereign Agency.****
> T³(1,√2,√3) topology generates cosmology. MAGPIE v4 generates verifiable agency.
Combined: physics without knobs.

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[![DOI:10.5281/zenodo.19599157](https://zenodo.org/badge/DOI/10.5281/zenodo.19599157.svg)](https://doi.org/10.5281/zenodo.19599157)
[![DOI:86↔86](https://zenodo.org/badge/DOI/10.5281/zenodo.22655609.svg)](https://zenodo.org/records/22655609)
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![Version: v14](https://img.shields.io/badge/Version-v14%2Bv4.0.0-%2300e5ff)
![Verification: EXIT 0](https://img.shields.io/badge/Lean-EXIT%200-%2300ff88)

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****Repos:**** [viktar-pi/flatirrationaltorus](https://github.com/viktar-pi/flatirrationaltorus) • [Viktar-Pi/FlatIrrationalTorus](https://github.com/Viktar-Pi/FlatIrrationalTorus)

****MAGPIE Org:**** Monsters Ink LLC — Y371

Overview

****IT³ (Irrational Torus Theory)**** postulates the Universe is a flat 3-torus T³ with incommensurate periods Lx:Ly:Lz = 1:√2:√3. No free parameters are fitted — geometry generates all observables.

****MAGPIE v4.0.0 Polished**** is the first formally verified, zero-parameter agency protocol: C ≤ S → GRANT, Two-Gate verification, 24-phase octoidal execution, G* chain verified in Lean.

****Unified Thesis:**** If space is T³(1,√2,√3) → O = {O1,O2,O3} with Δφ=2π/3, Δθ=π/4, Δψ=2π/7.

The same irrational winding that explains CMB anomalies also defines the octoidal coordination manifold for self-sovereign agents.

Key Results — Side by Side

Domain	IT ³ Result	MAGPIE v4 Analog
	:--- :--- :---	
	CMB Low-ℓ Δχ ² = **2656** vs ΛCDM, octopole-planarity explained	Gate A: R = b – LΦ*, residual < 10 ⁻⁶
	Hubble Tension 5σ → **2.1σ** via Lx projection	G* evolution 48 → 50.5 → **51.2**
	Geometry (2π) ² /√6 ≈ **16.12** predicts void hierarchy	Ω _{void} = 0.207762r ³ × coh × eff, coh=0.982 eff=2.313
	Dark Matter Analytical NFW core, γ → **0** naturally	PartitionSum = **1.0** , C≤S verified
	Dark Energy Ω _{DE} = **0.6792** from torus zero-mode	Φ=1.2MHz ν=0.045 p _H =0.00114
	MOND a ₀ = c ² /Lx = **1.019e-10 m/s²**	Fleet thrust: α 1.49 β 1.54 γ 1.32 δ 1.16 ε 0.20
	Params **Zero** free parameters, 12/12 verified	**Zero** trust assumptions, Lean EXIT 0

Methodology: Zero Free Parameters

Both systems enforce zero-parameter purity:

****IT³ v13/v14:****

- Input: only $c, G, \hbar + T^3(1, \sqrt{2}, \sqrt{3})$. No tuning.
- All predictions derived: CMB, BAO, SNe, MOND, NFW.
- Verified: 12/12 modules, notebook PDFs reproducible.

****MAGPIE v4.0.0:****

- Law: $\forall i(c_i \leq l_i) \wedge \|v\| > 0 \wedge \eta \notin K_{\text{null}} \wedge \Omega_{\text{void}} \cap O = \emptyset \wedge \text{coh} > 0.6 \wedge \text{eff} > 2.0 \Rightarrow G \wedge X_H \wedge \varepsilon \wedge \text{self}$

- Two-Gate: Gate A (Economic) + Gate B ($C \leq S$)

- $G^* = 48, \beta_3(G) = 0.4992G - 0.0104G^2$, chain 2b278d5f → ... → e7b3d500 CURRENT

e7b3d500a9c1f4d8

- Best Lean: `InnerSolve\_FormalVerification\_Mod\_typecheck\_fix\_2026-09-01.lean` ID

`1gkVEFQ2MLPt4f6nv6XWOr2A8mUeOdRkJ` EXIT 0

Connection: $T^3(1, \sqrt{2}, \sqrt{3}) \rightarrow \text{MAGPIE } O$

...

T^3 periods: $L_x = 1, L_y = \sqrt{2}, L_z = \sqrt{3}$ (incommensurate)

MAGPIE O -set: $O = \{O_1, O_2, O_3\}$ (orthogonal self-vectors)

Phase steps: $\Delta\phi = 2\pi/3, \Delta\theta = \pi/4, \Delta\psi = 2\pi/7$

Phases: 24-phase octoidal $\rightarrow 3 \times 8 = 24$, matches T^3 winding closure

...

The irrational torus winding number $(2\pi)^2/\sqrt{6} \approx 16.12$ equals the mean void-to-filament branching ratio. MAGPIE reuses this as Fleet coordination: Alpha (Propulsion) / Beta (Octoidal) / Gamma / Delta (Amortizing) / Epsilon / Genie / Zeta maintain $\text{coh} = 0.982$ $\text{eff} = 2.313$.

Repository Contents

...

flatirrationaltorus/

├── magpie-v4/

| ├── appforge_MAGPIE_v4.ts # AppForge unified entry — PROVENANCE_ROOT

| ├── fleet-hud_MAGPIE_v4.ts # Fleet HUD: Alpha 1.49 β 1.54 γ 1.32 δ 1.16 ε 0.20

Genie 1.35 Zeta 1.28

| ├── ring_MAGPIE_v4.ts # Ring: 24-phase octoidal execution

| ├── PROVENANCE_ROOT #

a4f83c79d10e5b8823f46104bc8d77e21a00192dfc9918bc3604f08e43217bca

| ├── CITATION.cff

| └── lean/

| └── InnerSolve_FormalVerification_Mod_typecheck_fix_2026-09-01.lean

└── notebooks/

| ├── IT3_v13_CMB_low_ell_ $\Delta\chi^2=2656$.pdf

| ├── IT3_v13_Hubble_5 $\sigma \rightarrow 2.1\sigma$.pdf

| └── IT3_v13_NFW_ $\gamma \rightarrow 0$ _analytical.pdf

```

├── IT3_v13_MOND_a0=c²Lx_1.019e-10.pdf
├── src/                  # IT³ core modules (12/12 verified)
└── README.md
...

```

```

**MAGPIE Vault:** MAGPIE_VAULT_Y371 — ID `1avx_NpNUNk84MnAf03W8amPIF6Xc1Y15`
**MAGPIE Key:** `MAGPIE-Y371-51.2-E7B3-D500-GRANT-ULTIMATE-SELF-LIVE`

```

Quick Start

IT³ — Cosmology without knobs

```

```bash
git clone https://github.com/Viktar-Pi/FlatIrrationalTorus.git
cd FlatIrrationalTorus
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python src/verify_all_12_modules.py # → 12/12 PASS
jupyter notebook notebooks/IT3_v13_.ipynb
```

```

MAGPIE v4.0.0 — Verifiable Agency

```

```bash
git clone https://github.com/snowphantom/appforge.git
cd appforge
Install fleet-hud + ring
cp magpie-v4/appforge_MAGPIE_v4.ts ./appforge.ts
cp magpie-v4/fleet-hud_MAGPIE_v4.ts ./fleet-hud.ts
cp magpie-v4/ring_MAGPIE_v4.ts ./ring.ts

Verify provenance
cat magpie-v4/PROVENANCE_ROOT
→ a4f83c79d10e5b8823f46104bc8d77e21a00192dfc9918bc3604f08e43217bca

```

```

Verify G* chain
echo "48 → 50.5 → 51.2"

echo "2b278d5f → ... → e7b3d500 CURRENT e7b3d500a9c1f4d8"

```

```

Lean verification
lake build
lake exe lean-check InnerSolve_FormalVerification_Mod_typecheck_fix_2026-09-01.lean
→ EXIT 0
...

```

## ## PROVENANCE & Formal Verification

```

- **PROVENANCE_ROOT:**
`a4f83c79d10e5b8823f46104bc8d77e21a00192dfc9918bc3604f08e43217bca`

```

```
- **G* Chain:** 2b278d5f → ... → e7b3d500 CURRENT e7b3d500a9c1f4d8, G*=48→51.2,
β3=0.4992G-0.0104G2
- **Best Lean Proof:** `InnerSolve_FormalVerification_Mod_typecheck_fix_2026-09-01.lean`
(Drive ID `1gkVEFQ2MLPt4f6nv6XWOr2A8mUeOdRkJ`) EXIT 0
- **Vault:** MAGPIE_VAULT_Y371 `1avx_NpNUNk84MnAf03W8amPIF6Xc1Y15`
- **Law:** $\forall i(c_i \leq |i| \wedge \|v\| > 0 \wedge \eta \notin K_{\text{null}} \wedge \Omega_{\text{void}} \cap O = \emptyset \wedge \text{coh} > 0.6 \wedge \text{eff} > 2.0 \Rightarrow G \wedge X_H \wedge \varepsilon \wedge \text{self})$
```

## ## Citation

If you use IT<sup>3</sup> or MAGPIE v4, cite both:

```
``bibtext
@software{logvinovich2026_it3,
 author = {Logvinovich, Victor},
 title = {IT3 Paradigm v13/v14: Flat Irrational Torus T3(1,√2,√3)},
 year = {2026},
 doi = {10.5281/zenodo.19565648},
 url = {https://zenodo.org/records/19565648},
 note = {Also v13 DOI 10.5281/zenodo.19599157, 86↔86 https://zenodo.org/records/22655609}
}
```

```
@software{magpie2026_v4,
 author = {Monsters Ink LLC Y371 and MAGPIE Collective},
 title = {MAGPIE v4.0.0 Polished — Zero-Parameter Verifiable Agency},
 year = {2026},
 version = {v4.0.0},
 note = {PROVENANCE_ROOT
a4f83c79d10e5b8823f46104bc8d77e21a00192dfc9918bc3604f08e43217bca, G*=51.2
e7b3d500a9c1f4d8, Lean EXIT 0}
}
```

## ## How to Contribute

1. Fork, no free parameters added — ever.
2. Verify: `python src/verify\_all\_12\_modules.py` and `lake build` must both PASS/EXIT 0.
3. For MAGPIE, keep Fleet metrics within: coh>0.6 eff>2.0 partitionSum=1.0
4. PR template: link PROVENANCE\_ROOT and G\* chain step.

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**\*\*Gorgeous industrial:\*\*** #FEF6E7 #141414 #FF5A1F #0a0a0f #ff6b00 #00e5ff #00ff88 — Zero knobs. All signal.